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NEET · BIOLOGY · CHAPTER-WISE PYQS

Anatomy of Flowering Plants

NEET Biology · Previous Year Questions with Answer Key

127	48	1988 – 2026	34	English
QUESTIONS	PAPERS COVERED	YEARS	OUT OF SYLLABUS	LANGUAGE

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MCQ (More than One Correct Answer)	1 question
Out of syllabus	34 questions
Answer key	at the end

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MCQ (Single Correct Answer) 92 questions

Q1. NEET 2026

The main function of bulliform cells in grasses is :

(A) to make the leaf impermeable to fungal spores.

(B) to transport water.

(C) to perform photosynthesis.

(D) to minimize water loss during water stress.

Open solution

Q2. NEET 2026

Match List-I with List-II :

List-I (Phase of cell cycle)		List-II (Activity)	
A.	Conjunctive tissue	I.	Specialised cells in the vicinity of guard cells
B.	Casparian strips	II.	Endodermal cells rich in starch
C.	Subsidiary cells	III.	Tissue between xylem and phloem
D.	Starch sheath	IV.	Endodermal cells with suberin deposition

Choose the correct answer from the options given below :

(A) A-IV, B-III, C-I, D-II

(B) A-III, B-IV, C-II, D-I

(C) A-III, B-IV, C-I, D-II

(D) A-IV, B-III, C-II, D-I

Open solution

Q3. NEET 2025

Find the statement that is NOT correct with regard to the structure of monocot stem.

(A) Vascular bundles are conjoint and closed.

(B) Phloem parenchyma is absent.

(C) Hypodermis is parenchymatous.

(D) Vascular bundles are scattered.

Open solution

Q4. NEET 2024 (Re-Examination)

Which of the following simple tissues are commonly found in the fruit walls of nuts and pulp of pear?

(A) Sclereids (B) Fibres (C) Parenchyma (D) Collenchyma

Open solution

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Q5. NEET 2024 (Re-Examination)

Which of the following helps in maintenance of the pressure gradient in sieve tubes?

(A) Albuminous cells
(B) Sieve cells
(C) Phloem parenchyma
(D) Companion cells

Open solution

Q6. NEET 2024 (Re-Examination)

Given below are two statements:

Statement I: In a dicotyledonous leaf, the adaxial epidermis generally bears more stomata than the abaxial epidermis.

Statement II: In a dicotyledonous leaf, the adaxially placed palisade parenchyma is made up of elongated cells, which are arranged vertically and parallel to each other.

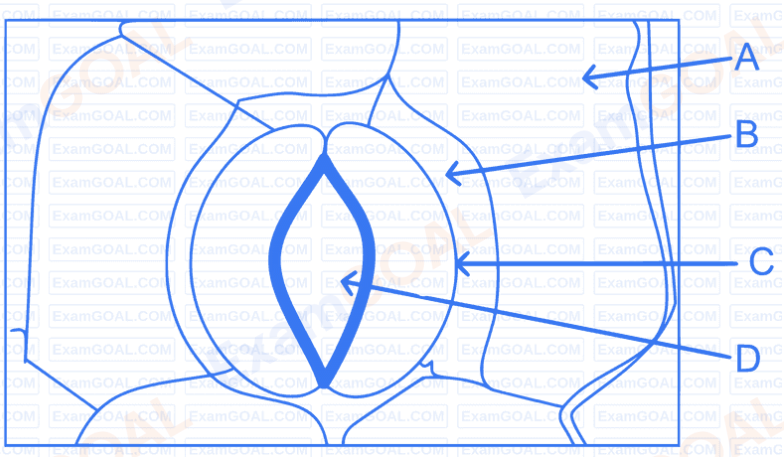
In the light of the above statements, choose the correct answer from the options given below:

(A) Statement I is true but Statement II is false
(B) Statement I is false but Statement II is true
(C) Both Statement I and Statement II are true
(D) Both Statement I and Statement II are false

Open solution

Q7. NEET 2024

In the given figure, which component has thin outer walls and highly thickened inner walls?



(A) C
(B) D
(C) A
(D) B

Open solution

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Q8. NEET 2024

Open solution

Given below are two statements:

Statement I : Parenchyma is living but collenchyma is dead tissue.

Statement II : Gymnosperms lack xylem vessels but presence of xylem vessels is the characteristic of angiosperms.

In the light of the above statements, choose the correct answer from the options given below:

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q9. NEET 2024

Open solution

Bulliform cells are responsible for

(A) Inward curling of leaves in monocots.

(B) Protecting the plant from salt stress.

(C) Increased photosynthesis in monocots.

(D) Providing large spaces for storage of sugars.

Q10. NEET 2023 Manipur

Open solution

The transverse section of a plant part showed polyarch, radial and exarch xylem, with endodermis and pericycle. The plant part is identified as :

(A) Monocot root **(B)** Dicot root **(C)** Dicot stem **(D)** Monocot stem

Q11. NEET 2023 Manipur

Open solution

Consider the following plant tissues :

(A) Axillary buds

(B) Fascicular vascular cambium

(C) Interfascicular cambium

(D) Cork cambium

(E) Intercalary meristem

Identify the lateral meristems among the above.

(A) (A), (C) and (D) only **(B)** (B), (C) and (D) only

(C) (A), (B), (C) and (E) only **(D)** (A), (B), (D) and (E) only

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Q12. NEET 2023

Given below are two statements :

Statement I : Endarch and exarch are the terms often used for describing the position of secondary xylem in the plant body.

Statement II : Exarch condition is the most common feature of the root system.

In the light of the above statements, choose the correct answer from the options given below:

(A) Both Statement I and Statement II are false

(B) Statement I is correct but Statement II is false

(C) Statement I is incorrect but Statement II is true

(D) Both Statement I and Statement II are true

Open solution

Q13. NEET 2022 Phase 2

Interfascicular cambium is present between

(A) Secondary xylem and secondary phloem

(B) Primary xylem and primary phloem

(C) Pericycle and endodermis

(D) Two vascular bundles

Open solution

Q14. NEET 2022 Phase 2

The type of tissue commonly found in the fruit wall of nuts is :

(A) Sclereid (B) Parenchyma (C) Collenchyma (D) Sclerenchyma

Open solution

Q15. NEET 2022 Phase 2

Initiation of lateral roots and vascular cambium during secondary growth takes place in cells of

(A) Pericycle (B) Epiblema (C) Cortex (D) Endodermis

Open solution

Q16. NEET 2022 Phase 1

Read the following statements about the vascular bundles:

(a) In roots, xylem and phloem in a vascular bundle are arranged in an alternate manner along the different radii.

(b) Conjoint closed vascular bundles do not possess cambium

(c) In open vascular bundles, cambium is present in between xylem and phloem

(d) The vascular bundles of dicotyledonous stem possess endarch protoxylem

(e) In monocotyledonous root, usually there are more than six xylem bundles present

Choose the correct answer from the options given below :

(A) (a), (b) and (d) only

(B) (a), (b), (c), (d) and (e) only

(C) (a), (b), (c) and (d) only

(D) (a), (c), (d) and (e) only

Open solution

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Q17. NEET 2021

Match List-I with List-II.

Open solution

	List-I		List-II
(a)	Cells with active cell division capacity	(i)	Vascular tissues
(b)	Tissue having all cells similar in structure and function	(ii)	Meristematic tissue
(c)	Tissue having different types of cells	(iii)	Sclereids
(d)	Dead cells with highly thickened walls and narrow lumen	(iv)	Simple tissue

Select the correct answer from the options given below.

(A)

(a)	(b)	(c)	(d)
(iii)	(ii)	(iv)	(i)

(B)

(a)	(b)	(c)	(d)
(ii)	(iv)	(i)	(iii)

(C)

(a)	(b)	(c)	(d)
(iv)	(iii)	(ii)	(i)

(D)

(a)	(b)	(c)	(d)
(i)	(ii)	(iii)	(iv)

Q18. NEET 2021

Select the correct pair.

Open solution

(A) Loose parenchyma cells rupturing the epidermis and forming a lens shaped opening in bark - Spongy parenchyma

(B) Large colorless empty cells in the epidermis of grass leaves - Subsidiary cells

(C) In dicot leaves, vascular bundles are surrounded by large thick-walled cells - Conjunctive tissue

(D) Cells of medullary rays that form part of cambial ring - Interfascicular cambium

Q19. NEET 2020 Phase 2

Large, empty colourless cells of the adaxial epidermis along the veins of grass leaves are

Open solution

(A) Bulliform cells

(B) Lenticels

(C) Guard cells

(D) Bundle sheath cells

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Q20. NEET 2020 Phase 1

The transverse section of a plant shows following anatomical features :

(i) Large number of scattered vascular bundles surrounded by bundle sheath
(ii) Large conspicuous parenchymatous ground tissue
(iii) Vascular bundles conjoint and closed
(iv) Phloem parenchyma absent

Identify the category of plant and its part:

(A) Monocotyledonous root
(B) Dicotyledonous stem
(C) Dicotyledonous root
(D) Monocotyledonous stem

Open solution

Q21. NEET 2019 (Odisha)

Regeneration of damaged growing grass following grazing is largely due to :

(A) Secondary meristem
(B) Lateral meristem
(C) Apical meristem
(D) Intercalary meristem

Open solution

Q22. NEET 2019

Grass leaves curl inward during very dry weather. Select the most appropriate reason from the following :

(A) Shrinkage of air spaces in spongy mesophyll
(B) Tyloses in vessels
(C) Closure of stomata
(D) Flaccidity of bulliform cells

Open solution

Q23. NEET 2019

Phloem in gymnosperms lacks :

(A) Companion cells only
(B) Both sieve tubes and companion cells
(C) Albuminous cells and sieve cells
(D) Sieve tubes only

Open solution

Q24. NEET 2019

What is the direction of movement of sugars in phloem?

(A) Downward
(B) Bi-directional
(C) Non-multidirectional
(D) Upward

Open solution

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Q25. NEET 2018

Stomata in grass leaf are

(A) kidney-shaped (B) barrel-shaped (C) dumb-bell shaped (D) rectangular

Open solution

Q26. NEET 2018

Casparian strips occur in

(A) pericycle (B) epidermis (C) endodermis. (D) cortex

Open solution

Q27. NEET 2017

Which of the following is made up of dead cells ?

(A) Collenchyma (B) Xylem parenchyma
(C) Phellem (D) Phloem

Open solution

Q28. NEET 2016 Phase 2

Cortex is the region found between -

(A) endodermis and pith (B) endodermis and vascular bundle
(C) epidermis and stele (D) pericycle and endodermis

Open solution

Q29. NEET 2016 Phase 1

Specialised epidermal cells surrounding the guard cells are called -

(A) Lenticels (B) Complementary cells
(C) Bulliform cells (D) Subsidiary cells

Open solution

Q30. AIPMT 2015 Cancelled Paper

A major characteristic of the monocot root is the presence of :

(A) Vasculature without cambium
(B) Cambium sandwiched between phloem and xylem along the radius
(C) Open vascular bundles
(D) Scattered vascular bundles

Open solution

Q31. AIPMT 2015 Cancelled Paper

Vascular bundles in monocotyledons are considered closed because

(A) There are no vessels with perforations (B) Xylem is surrounded all around by phloem
(C) A bundle sheath surrounds each bundle (D) Cambium is absent

Open solution

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Q32. AIPMT 2014

You are given a fairly old piece of dicot stem and a dicot root. Which of the following anatomical structures will you use to distinguish between the two?

(A) Secondary xylem
(B) Protoxylem
(C) Secondary phloem
(D) Cortical cells

Open solution

Q33. AIPMT 2014

Tracheids differ from other tracheary elements in :

(A) being imperforate
(B) having casparian strips
(C) lacking nucleus
(D) being lignified

Open solution

Q34. NEET 2013 (Karnataka)

Meristematic tissue responsible for increase in girth of tree trunk is :

(A) Phellogen
(B) Intercalary meristem
(C) Apical meristem
(D) Lateral meristem

Open solution

Q35. NEET 2013 (Karnataka)

Which of the following statements is not true for stomatal apparatus ?

(A) Stomata are involved in gaseous exchange
(B) Inner walls of guard cells are thick
(C) Guard cells invariably possess chloroplasts and mitochondria
(D) Guard cells are always surrounded by subsidiary cells

Open solution

Q36. AIPMT 2012 Mains

As compared to a dicot root, a monocot root has

(A) More abundant secondary xylem
(B) Relatively thicker periderm
(C) Many xylem bundles
(D) Inconspicuous annual rings

Open solution

Q37. AIPMT 2012 Prelims

Gymnosperms are also called soft wood spermatophytes because they lack

(A) Cambium
(B) Thick - walled tracheids
(C) Phloem fibres
(D) Xylem fibres

Open solution

Q38. AIPMT 2012 Prelims

Water containing cavities in vascular bundles are found in

(A) Maize
(B) Cycas
(C) Sunflower
(D) Pinus

Open solution

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Q39. AIPMT 2012 Prelims

Closed vascular bundles lack

(A) Ground tissue
(B) Conjunctive tissue
(C) Cambium
(D) Pith

Open solution

Q40. AIPMT 2012 Prelims

Companion cells are closely associated with

(A) Sieve elements
(B) Trichomes
(C) Vessel elements
(D) Guard cells

Open solution

Q41. AIPMT 2011 Mains

Some vascular bundles are described as open because these :

(A) are capable of producing secondary xylem and phloem
(B) are not surrounded by pericycle
(C) possess conjunctive tissue between xylem and phloem
(D) are surrounded by pericycle but no endodermis

Open solution

Q42. AIPMT 2011 Mains

Function of companion cells is :

(A) Providing water to phloem
(B) Loading of sucrose into sieve elements
(C) Loading of sucrose into sieve elements by passive transport
(D) Providing energy to sieve elements for active transport

Open solution

Q43. AIPMT 2011 Prelims

Ground tissue includes :

(A) All tissues external to endodermis
(B) Epidermis and cortex
(C) All tissues internal to endodermis
(D) All tissues except epidermis and vascular bundles

Open solution

Q44. AIPMT 2010 Mains

Transport of food material in higher plants takes place through -

(A) Sieve elements
(B) Transfusion tissue
(C) Companion cells
(D) Tracheids

Open solution

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Q45. AIPMT 2010 Prelims

Heartwood differs from sapwood in –

(A) Having dead and non-conducting elements
(B) Being susceptible to pests and pathogens
(C) Presence of rays and fibres
(D) Abscence of vessels and parenchyma

Open solution

Q46. AIPMT 2010 Prelims

Which one of the folliwng is not a lateral meristem ?

(A) Interfascicular cambium
(B) Intercalary meristem
(C) Intrafascicular cambium
(D) Phellogen

Open solution

Q47. AIPMT 2010 Prelims

The chief water conducting elements of xylem in gymnosperms are :

(A) Fibres
(B) Transfusion tissue
(C) Vessels
(D) Tracheids

Open solution

Q48. AIPMT 2009

In barley stem vascular bundles are -

(A) Closed and scattered
(B) Closed and radial
(C) Open and in a ring
(D) Open and scattered

Open solution

Q49. AIPMT 2009

Palisade parenchyma is absent in leaves of -

(A) Mustard
(B) Sorghum
(C) Soybean
(D) Gram

Open solution

Q50. AIPMT 2009

Anatomically fairly old dicotyledonous root is distinguished from the dicotyledonous stem by :

(A) Absence of secondary phloem
(B) Presence of cortex
(C) Position of protoxylem
(D) Absence of secondary xylem

Open solution

Q51. AIPMT 2008

Vascular tissues in flowering plants develop from:

(A) plerome
(B) periblem
(C) phellogen
(D) dermatogen

Open solution

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Q52. AIPMT 2008

The length of different internodes in a culm of sugarcane is variable because of:

(A) shoot apical meristem

(B) position of axillary buds

(C) size of leaf lamina at the node below each internode

(D) intercalary meristem

Open solution

Q53. AIPMT 2007

Passage cells are thin-walled cells found in:-

(A) Endodermis of roots facilitating rapid transport of water from cortex to pericycle

(B) Testa of seeds to enable emergence of growing embryonic axis during seed germination.

(C) Phloem elements that serve as entry points for substances for transport to other plant parts

(D) Central region of style through which the pollen tube grows towards the ovary.

Open solution

Q54. AIPMT 2006

A common structural feature of vessel elements and sieve tube elements is-

(A) Enucleate condition

(B) Presence of p-protein

(C) Thick secondary walls

(D) Pores on lateral wall

Open solution

Q55. AIPMT 2003

The apical meristem of the root is present

(A) In all the roots

(B) Only in tap roots

(C) Only in adventitious roots

(D) Only in radicals

Open solution

Q56. AIPMT 2003

In which one of the following nitrogen is not a constituent ?

(A) Pepsin

(B) Invertase

(C) Idioblast

(D) Bacteriochlorophyll

Open solution

Q57. AIPMT 2003

The aleurone layer in maize grain is specially rich in :-

(A) Lipids

(B) Proteins

(C) Auxins

(D) Starch

Open solution

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Q58. AIPMT 2003

Chlorenchyma is known to develop in the : -

(A) Mycelium of a green mould such as Aspergillus

(B) Pollen tube of Pinus

(C) Cytoplasm of Chlorella

(D) Spore capsule of a moss

Open solution

Q59. AIPMT 2003

Diffuse porous wood is characteristic of those plants which are growing in : -

(A) Tropics

(B) Temperate climate

(C) Alpine region

(D) Cold winter regions

Open solution

Q60. AIPMT 2003

The cells of the quiescent centre are characterised by : -

(A) Having light cytoplasm and small nuclei

(B) Dividing regularly to add to tunica

(C) Having dense cytoplasm and prominent nuclei

(D) Dividing regularly to add to the corpus

Open solution

Q61. AIPMT 2002

Which of the following statement is true ?

(A) Tracheids are unicellular and with wide lumen

(B) Tracheids are multicellular and with narrow lumen

(C) Vessels are multicellular and with wide lumen

(D) Vessels are unicellular and with narrow lumen

Open solution

Q62. AIPMT 2002

Axillary bud and terminal bud derived from the activity of : -

(A) Apical meristem

(B) Parenchyma

(C) Lateral meristem

(D) Intercalary meristem

Open solution

Q63. AIPMT 2002

Four radial vascular bundle are found in : -

(A) Monocot stem

(B) Dicot stem

(C) Dicot root

(D) Monocot root

Open solution

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Q64. AIPMT 2002

Vessels are found in : -

(A) Most of the angiosperm and few gymnosperms

(B) All angiosperms and some gymnosperm

(C) All pteridophyta

(D) All angiosperms, all gymnosperms and some pteridophyta

Open solution

Q65. AIPMT 2001

In which of the following plant sunken stomata are found?

(A) *Nerium* (B) *Hydilla* (C) *Mango* (D) Guava

Open solution

Q66. AIPMT 2001

In plants inulin and pectin are

(A) Excretory material (B) Reserved material

(C) Insect attracting material (D) Wastes

Open solution

Q67. AIPMT 2000

What happens in plants during vascularisation ?

(A) Formation of procambium, primary phloem and xylem simultaneousl

(B) Differentiation of procambium followed by the formation of secondary xylem

(C) Differentiation of procambium, formation of primary phloem followed by formation of primary xylem

(D) Differentiation of procambium followed by the formation of primary phloem and xylem simultaneously

Open solution

Q68. AIPMT 1999

Casparian strips are found in :

(A) hypodermis (B) endodermis (C) periderm (D) epidermis

Open solution

Q69. AIPMT 1997

At maturity, which of the following is non-nucleated?

(A) Palisade cell (B) Cortical cell (C) Sieve cell (D) Companion cell

Open solution

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Q70. AIPMT 1996

Which of the following is not true about 'sclereids'?

(A) These are groups of living cells.

(B) These are found in nut shells, guava pulp, pear.

(C) These are also called stone cells.

(D) These are form of sclerenchyma with fibres.

Open solution

Q71. AIPMT 1994

Casparian strip occurs in a

(A) endodermis (B) exodermis (C) pericycle (D) epidermis

Open solution

Q72. AIPMT 1994

Where do the casparian bands occur?

(A) Epidermis (B) Endodermis (C) Pericycle (D) Phloem

Open solution

Q73. AIPMT 1993

Which of the following plant cells will show totipotency?

(A) Sieve tubes (B) Xylem vessels (C) Meristem (D) Cork cells

Open solution

Q74. AIPMT 1993

Bordered pits are found in

(A) sieve cells (B) vessel wall (C) companion cells (D) sieve tube wall

Open solution

Q75. AIPMT 1992

A bicollateral vascular bundle is characterised by

(A) phloem being sandwiched between xylem

(B) transverse splitting of vascular bundle

(C) longitudinal splitting of vascular bundle

(D) xylem being sandwiched between phloem

Open solution

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Q76. AIPMT 1992

What is true about a monocot leaf?

(A) Reticulate venation

(B) Absence of bulliform cells from epidermis

(C) Mesophyll not differentiated into palisade and spongy tissues

(D) Well differentiated mesophyll

Open solution

Q77. AIPMT 1991

An organised and differentiated cellular structure having cytoplasm but no nucleus is

(A) vessels

(B) xylem parenchyma

(C) sieve tubes

(D) tracheids

Open solution

Q78. AIPMT 1991

Angular collenchyma occurs in

(A) *Cucurbita*

(B) *Helianthus*

(C) *Althaea*

(D) *Salvia*

Open solution

Q79. AIPMT 1990

Where do the casparian bands occur?

(A) Epidermis

(B) Endodermis

(C) Pericycle

(D) Phloem

Open solution

Q80. AIPMT 1990

What is true about a monocot leaf?

(A) Reticulate venation

(B) Absence of bulliform cells from epidermis

(C) Mesophyll not differentiated into palisade and spongy tissues

(D) Well differentiated mesophyll

Open solution

Q81. AIPMT 1990

For union between stock and scion in grafting which one is the first to occur

(A) formation of callus

(B) production of plasmodesmata

(C) differentiation of new vascular tissues

(D) regeneration of cortex and epidermis

Open solution

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Q82. AIPMT 1990

Pericycle of roots produces

(A) mechanical support
(B) lateral roots
(C) vascular bundles
(D) adventitious buds

Open solution

Q83. AIPMT 1990

Collenchyma occurs in

(A) herbaceous climbers
(B) woody climbers
(C) climbing stems
(D) water plants

Open solution

Q84. AIPMT 1990

Collenchyma occurs in the stem and petioles of

(A) xerophytes
(B) monocots
(C) dicot herbs
(D) hydrophytes

Open solution

Q85. AIPMT 1990

Monocot leaves possess

(A) intercalary meristem
(B) lateral meristem
(C) apical meristem
(D) mass meristem

Open solution

Q86. AIPMT 1990

Cork cambium and vascular cambium are

(A) parts of secondary xylem and phloem
(B) parts of pericycle
(C) lateral meristem
(D) apical meristem

Open solution

Q87. AIPMT 1989

Sieve tubes are suited for translocation of food because they possess

(A) bordered pits
(B) no end walls
(C) broader lumen and perforated cross walls
(D) no protoplasm

Open solution

Q88. AIPMT 1989

Death of protoplasm is a pre-requisite for a vital function like

(A) transport of sap
(B) transport of food
(C) absorption of water
(D) gaseous exchange

Open solution

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Q89. AIPMT 1988

Organisation of stem apex into corpus and tunica is determined mainly by

(A) planes of cell division
(B) regions of meristematic activity
(C) rate of cell growth
(D) rate of shoot tip growth

Open solution

Q90. AIPMT 1988

Pith and cortex do not differentiate in

(A) monocot stem
(B) dicot stem
(C) monocot root
(D) dicot root

Open solution

Q91. AIPMT 1988

Which meristem helps in increasing girth?

(A) Lateral meristem
(B) Intercalary meristem
(C) Primary meristem
(D) Apical meristem

Open solution

Q92. AIPMT 1988

Tunica corpus theory is connected with

(A) root apex
(B) root cap
(C) shoot apex
(D) secondary growth

Open solution

MCQ (More than One Correct Answer) 1 questions

Q93. AIPMT 2009

The annular and spirally thickened conducting elements generally develop in the protoxylem when the root or stem is -

(A) Maturing
(B) Widening
(C) Elongating
(D) Differentiating

Open solution

Out of syllabus 34 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

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Q94. NEET 2023 Manipur · MCQ (Single Correct Answer)

Open solution

Consider the following tissues in the stelar region of a stem showing secondary growth.

(A) Primary xylem

(B) Secondary xylem

(C) Primary phloem

(D) Secondary phloem

Arrange these in the correct sequence of their position from pith towards cortex.

(A) (A), (B), (D), (C)

(B) (B), (A), (C), (D)

(C) (A), (B), (C), (D)

(D) (B), (A), (D), (C)

Q95. NEET 2023 · MCQ (Single Correct Answer)

Open solution

Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R :

Assertion A : Late wood has fewer xylary elements with narrow vessels.

Reason R : Cambium is less active in winters.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both A and R are true but R is NOT the correct explanation of A

(B) A is true but R is false

(C) A is false but R is true

(D) Both A and R are true and R is the correct explanation of A

Q96. NEET 2023 · MCQ (Single Correct Answer)

Open solution

Identify the correct statements:

A. Lenticels are the lens-shaped openings permitting the exchange of gases.

B. Bark formed early in the season is called hard bark.

C. Bark is a technical term that refers to all tissues exterior to vascular cambium.

D. Bark refers to periderm and secondary phloem.

E. Phellogen is single-layered in thickness.

Choose the correct answer from the options given below:

(A) A and D only **(B)** A, B and D only **(C)** B and C only **(D)** B, C and E only

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Q97. NEET 2022 Phase 1 · MCQ (Single Correct Answer)

Open solution

In old trees the greater part of secondary xylem is dark brown and resistant to insect attack due to :

(a) secretion of secondary metabolities and their deposition in the lumen of vessels.

(b) deposition of organic compounds like tannins and resins in the central layers of stem.

(c) deposition of suberin and aromatic substances in the outer layer of stem.

(d) deposition of tannins, gum, resin and aromatic substances in the peripheral layers of stem.

(e) presence of parenchyma cells, functionally active xylem elements and essential oils.

Choose the correct answer from the options given below :

(A) (a) and (b) only **(B)** (c) and (d) only **(C)** (d) and (e) only **(D)** (b) and (d) only

Q98. NEET 2022 Phase 1 · MCQ (Single Correct Answer)

Open solution

The anatomy of springwood shows some peculiar features. Identify the correct set of statements about springwood.

(a) It is also called as the earlywood

(b) In spring season cambium produces xylem elements with narrow vessels

(c) It is lighter in colour

(d) The springwood along with autumn wood shows alternate concentric rings forming annual rings

(e) It has lower density

Choose the correct answer from the options given below:

(A) (a), (b), (d) and (e) only **(B)** (a), (c), (d) and (e) only

(C) (a), (b) and (d) only **(D)** (c), (d) and (e) only

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Q99. NEET 2021 · MCQ (Single Correct Answer)

Open solution

Match List-I with List-II.

	List-I		List-II
(a)	Lenticels	(i)	Phellogen
(b)	Cork cambium	(ii)	Suberin deposition
(c)	Secondary cortex	(iii)	Exchange of gases
(d)	Cork	(iv)	Phelloderm

Choose the correct answer from the options given below.

(A)

(a)	(b)	(c)	(d)
(iv)	(ii)	(i)	(iii)

(B)

(a)	(b)	(c)	(d)
(iv)	(i)	(iii)	(ii)

(C)

(a)	(b)	(c)	(d)
(iii)	(i)	(iv)	(ii)

(D)

(a)	(b)	(c)	(d)
(ii)	(iii)	(iv)	(i)

Q100. NEET 2020 Phase 2 · MCQ (Single Correct Answer)

Open solution

Which of the following statements about cork cambium is **incorrect** ?

(A) It is a couple of layers thick

(B) It forms secondary cortex on its outerside

(C) It forms a part of periderm

(D) It is responsible for the formation of lenticels

Q101. NEET 2020 Phase 1 · MCQ (Single Correct Answer)

Open solution

Identify the incorrect statement.

(A) Sapwood is involved in conduction of water and minerals from root to leaf

(B) Sapwood is the innermost secondary xylem and is lighter in colour

(C) Due to deposition of tannins, resins, oils etc., heart wood is dark in colour

(D) Heart wood does not conduct water but gives mechanical support

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Q102. NEET 2019 (Odisha) · MCQ (Single Correct Answer)

In the dicot root the vascular cambium originates from :

(A) Intrafascicular and interfascicular tissue in a ring

(B) Tissue located below the phloem bundles and a portion of pericycle tissue above protoxylem.

(C) Cortical region

(D) Parenchyma between endodermis and pericycle

Open solution

Q103. NEET 2019 · MCQ (Single Correct Answer)

Which of the statements given below is not true about formation of Annual Rings in trees?

(A) Activity of cambium depends upon variation in climate.

(B) Annula ring are not prominent in trees of temperate region.

(C) Annula ring is a combination of spring wood and autumn wood produced in a year.

(D) Differential activity of cambium causes light and dark bands of tissue-early and late wood respectively

Open solution

Q104. NEET 2018 · MCQ (Single Correct Answer)

Secondary xylem and phloem in dicot stem are produced by

(A) vascular cambium (B) axillary meristems

(C) apical meristems (D) phellogen

Open solution

Q105. NEET 2018 · MCQ (Single Correct Answer)

Plants having little or no secondary growth are

(A) deciduous angiosperms (B) grasses

(C) cycads (D) conifers

Open solution

Q106. NEET 2017 · MCQ (Single Correct Answer)

The vascular cambium normally gives rise to :

(A) Primary phloem (B) Secondary xylem (C) Periderm (D) Phelloderm

Open solution

Q107. NEET 2017 · MCQ (Single Correct Answer)

Identify the wrong statement in context of heartwood :

(A) It is highly durable.

(B) It comprises dead elements with highly lignified walls

(C) It conducts water and minerals efficiently

(D) Organic compounds are deposited in it

Open solution

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Q108. NEET 2016 Phase 2 · MCQ (Single Correct Answer)

The balloon-shaped structures called tyloses

(A) are extensions of xylem parenchyma cells into vessels

(B) originate in the lumen of vessels

(C) are linked to the ascent of sap through xylem vessels

(D) characterize the sapwood

Open solution

Q109. AIPMT 2015 · MCQ (Single Correct Answer)

Read the different components from (1) to (4) in the list given below and tell the correct order of the components with reference to their arrangement from outer side to inner side in a woody dicot stem :

(1) Secondary cortex
(2) Wood
(3) Secondary phloem
(4) Phellem

The correct order is :

(A) (3), (4), (2), (1)

(B) (4), (1), (3), (2)

(C) (4), (3), (1), (2)

(D) (1), (2), (4), (3)

Open solution

Q110. NEET 2013 · MCQ (Single Correct Answer)

Lenticels are involved in

(A) Food transport

(B) Photosynthesis

(C) Transpiration

(D) Gaseous exchange

Open solution

Q111. NEET 2013 · MCQ (Single Correct Answer)

Age of a tree can be estimated by :

(A) Diameter of its heartwood

(B) Number of annual rings

(C) Its height and girth

(D) Biomass

Open solution

Q112. NEET 2013 · MCQ (Single Correct Answer)

Interfascicular cambium develops from the cells of :

(A) Endodermis

(B) Pericycle

(C) Medullary rays

(D) Xylem parenchyma

Open solution

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Q113. AIPMT 2012 Prelims · MCQ (Single Correct Answer)

The common bottle cork is a product of

(A) Phellogen
(B) Xylem
(C) Vascular Cambium
(D) Dermatogen

Open solution

Q114. AIPMT 2011 Prelims · MCQ (Single Correct Answer)

The cork cambium, cork and secondary cortex are collectively called :

(A) Periderm
(B) Phellem
(C) Phellogen
(D) Phelloderm

Open solution

Q115. AIPMT 2007 · MCQ (Single Correct Answer)

For a critical study of secondary growth in plants, which one of the following pairs is suitable ?

(A) Deodar and fern
(B) Wheat and maiden hair fern
(C) Sugarcane and sunflower
(D) Teak and pine

Open solution

Q116. AIPMT 2005 · MCQ (Single Correct Answer)

In a woody dicotyledonous tree, which of the following parts will mainly consist of primary tissues ?

(A) Shoot tips and root tips
(B) Flowers, fruit and leaves
(C) Stem and root
(D) All parts

Open solution

Q117. AIPMT 1998 · MCQ (Single Correct Answer)

Which of the following meristems is responsible for extrastelar secondary growth in dicotyledonous stem?

(A) Phellogen
(B) Intra-fascicular cambium
(C) Inter-fascicular cambium
(D) Inter-calary meristem

Open solution

Q118. AIPMT 1998 · MCQ (Single Correct Answer)

The periderm includes

(A) cork
(B) cambium
(C) secondary phloem
(D) all of these

Open solution

Q119. AIPMT 1994 · MCQ (Single Correct Answer)

As a tree grows older, which of the following increases more rapidly in thickness?

(A) Heart wood
(B) Sapwood
(C) Phloem
(D) Cortex

Open solution

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Q120. AIPMT 1993 · MCQ (Single Correct Answer)

Periderm is produced by

(A) vascular cambium
(B) fascicular cambium
(C) phellogen
(D) intrafascicular cambium

Open solution

Q121. AIPMT 1993 · MCQ (Single Correct Answer)

A narrow layer of thin walled cells found between phloem/bark and wood of a dicot is

(A) cork cambium
(B) vascular cambium
(C) endodermis
(D) pericycle

Open solution

Q122. AIPMT 1993 · MCQ (Single Correct Answer)

Which exposed wood will decay faster?

(A) Sapwood
(B) Softwood
(C) Wood with lot of fibres
(D) Heartwood

Open solution

Q123. AIPMT 1993 · MCQ (Single Correct Answer)

Abnormal/anomalous secondary growth occurs in

(A) *Dracaena*
(B) ginger
(C) wheat
(D) sunflower

Open solution

Q124. AIPMT 1992 · MCQ (Single Correct Answer)

Vascular cambium produces

(A) primary xylem and primary phloem
(B) secondary xylem and secondary phloem
(C) primary xylem and secondary phloem
(D) secondary xylem and primary phloem

Open solution

Q125. AIPMT 1990 · MCQ (Single Correct Answer)

Vascular cambium produces

(A) primary xylem and primary phloem
(B) secondary xylem and secondary phloem
(C) primary xylem and secondary phloem
(D) secondary xylem and primary phloem

Open solution

Q126. AIPMT 1989 · MCQ (Single Correct Answer)

Out of diffuse porous and ring porous woods, which is correct?

(A) Ring porous wood carries more water for short period.
(B) Diffuse porous wood carries more water.
(C) Ring porous wood carries more water when need is higher.
(D) Diffuse porous wood is less specialised but conducts water rapidly throughout.

Open solution

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Q127. AIPMT 1988 · MCQ (Single Correct Answer)

Cork is formed from

(A) cork cambium (phellogen)

(B) vascular cambium

(C) phloem

(D) xylem

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **D**	2. **C**	3. **C**	4. **A**	5. **D**
6. **B**	7. **A**	8. **D**	9. **A**	10. **A**
11. **B**	12. **C**	13. **D**	14. **A**	15. **A**
16. **B**	17. **B**	18. **D**	19. **A**	20. **D**
21. **D**	22. **D**	23. **B**	24. **B**	25. **C**
26. **C**	27. **C**	28. **C**	29. **D**	30. **A**
31. **D**	32. **B**	33. **A**	34. **D**	35. **D**
36. **C**	37. **D**	38. **A**	39. **C**	40. **A**
41. **A**	42. **B**	43. **D**	44. **A**	45. **A**
46. **B**	47. **D**	48. **A**	49. **B**	50. **C**
51. **A**	52. **D**	53. **A**	54. **A**	55. **A**
56. **C**	57. **B**	58. **D**	59. **A**	60. **A**
61. **C**	62. **A**	63. **C**	64. **A**	65. **A**
66. **B**	67. **D**	68. **B**	69. **C**	70. **A**
71. **A**	72. **B**	73. **C**	74. **B**	75. **D**
76. **C**	77. **C**	78. **A**	79. **B**	80. **C**
81. **A**	82. **B**	83. **C**	84. **C**	85. **A**
86. **C**	87. **C**	88. **A**	89. **A**	90. **A**
91. **A**	92. **C**			

MCQ (More than One Correct Answer)

93. **C, D**

Out of syllabus

94. **A**	95. **D**	96. **A**	97. **A**	98. **B**
99. **C**	100. **B**	101. **B**	102. **B**	103. **B**
104. **A**	105. **B**	106. **B**	107. **C**	108. **A**
109. **B**	110. **D**	111. **B**	112. **C**	113. **A**
114. **A**	115. **D**	116. **B**	117. **A**	118. **D**
119. **A**	120. **C**	121. **B**	122. **A**	123. **A**
124. **B**	125. **B**	126. **C**	127. **A**	

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